

Task 1:

Literature Review: Auto insurance risk assessment requires comprehensive analysis of multiple data points to accurately predict claim likelihood and severity. This literature review examines key sources of domain knowledge, identifies critical risk factors, and justifies the necessity of a data warehouse solution for effective risk analysis in the auto insurance industry.

a. Sources Used for Information Gathering

The following sources were used to gain domain knowledge on auto insurance and risk assessment:

1. Insurance Bureau of Canada (IBC)

This source provided foundational information about types of auto insurance coverage in Canada, how premiums are calculated, and what drivers need to know when purchasing insurance.

2. Ratehub

This site offered insights into the factors that influence auto insurance rates, including vehicle type, driver demographics, and driving history. It also compared quotes from different insurance providers.

3. Desjardins Insurance

This source detailed the various auto insurance coverage options, such as liability, comprehensive, and accident benefits. It also highlighted optional coverages and their benefits.

4. OTLA Blog – FSCO Rate Filings & Transparency

This article discussed rate filing practices, profit transparency, and regulatory concerns in the auto insurance industry in Ontario. It emphasized the importance of data in understanding insurer profits and customer costs.

5. TD Insurance

This resource provided information on risk assessment methodologies, claims processing procedures, and how different vehicle safety features can impact premium calculations. Their documentation also offered insights into emerging trends such as telematics and usage-based insurance programs that are transforming traditional risk assessment models.

b. Factors that may affect the risk of auto insurance: Auto insurance premiums are determined by evaluating various factors that influence the likelihood and potential cost of

claims. These factors can be broadly categorized into driver-related, vehicle-related, and usage-related considerations.

Driver-Related Factors:

- **Age:** Younger drivers, particularly teenagers, are statistically more prone to accidents, leading to higher premiums. Conversely, drivers over 25 often see reduced rates due to a perceived decrease in risk.
- **Gender and Marital Status:** Insurance companies may consider gender and marital status when assessing risk profiles. For instance, some data suggest that married individuals may have fewer accidents, potentially resulting in lower premiums.
- **Driving History:** A record of traffic violations or at-fault accidents can increase premiums, as it indicates a higher risk. Conversely, a clean driving record can lead to discounts.
- **Credit History:** In some regions, insurers use credit scores to assess financial responsibility, with lower scores potentially leading to higher premiums.

Vehicle-Related Factors:

- **Make and Model:** High-performance or luxury vehicles often cost more to insure due to increased repair or replacement costs and a higher likelihood of theft.
Vehicle Age and Value: Newer, more expensive cars may lead to higher premiums because of their replacement value, while older vehicles might be cheaper to insure.
- **Safety Features:** According to [isure.ca](https://www.isure.ca), vehicles equipped with advanced safety features may qualify for discounts, as they can reduce the severity of injuries and damage in accidents.

Usage-Related Factors:

- **Annual Mileage:** Depending on how often you drive, you may be at an increased risk of being involved in an accident. And accidents can impact your insurance premium.
- **Purpose of Use:** Using a vehicle for business purposes may result in higher premiums compared to personal use, due to increased exposure to risk.

External Factors:

- **Location:** Areas with higher traffic density, crime rates, or accident frequencies can lead to increased premiums due to the elevated risk of incidents.
- **Climate and Weather Conditions:** Regions prone to severe weather events may experience higher insurance costs due to the increased likelihood of weather-related claims.

Understanding these factors enables insurers to assess risk more accurately and set premiums that reflect the likelihood and potential cost of claims.

c. Justify why a data warehouse is necessary to collect and store analytical data for risk analysis of auto insurance.

In the auto insurance industry, the ability to effectively analyze risk is paramount. A data warehouse serves as an essential infrastructure for this purpose, offering several key advantages:

1. Centralized Data Integration: Insurance companies accumulate vast amounts of data from diverse sources, including policy details, claims records, customer demographics, and external data like traffic reports and weather conditions. A data warehouse consolidates this disparate information into a unified repository, facilitating comprehensive analysis and reporting.

2. Enhanced Analytical Capabilities: With data centralized, insurers can employ advanced analytics, such as predictive modeling and machine learning, to identify patterns and trends. This enables more accurate risk assessments, allowing for better underwriting decisions and personalized pricing strategies.

3. Improved Data Quality and Consistency: A data warehouse enforces standardized data formats and validation rules, ensuring that the information used for analysis is accurate and consistent. This reliability is crucial for making informed business decisions and maintaining regulatory compliance.

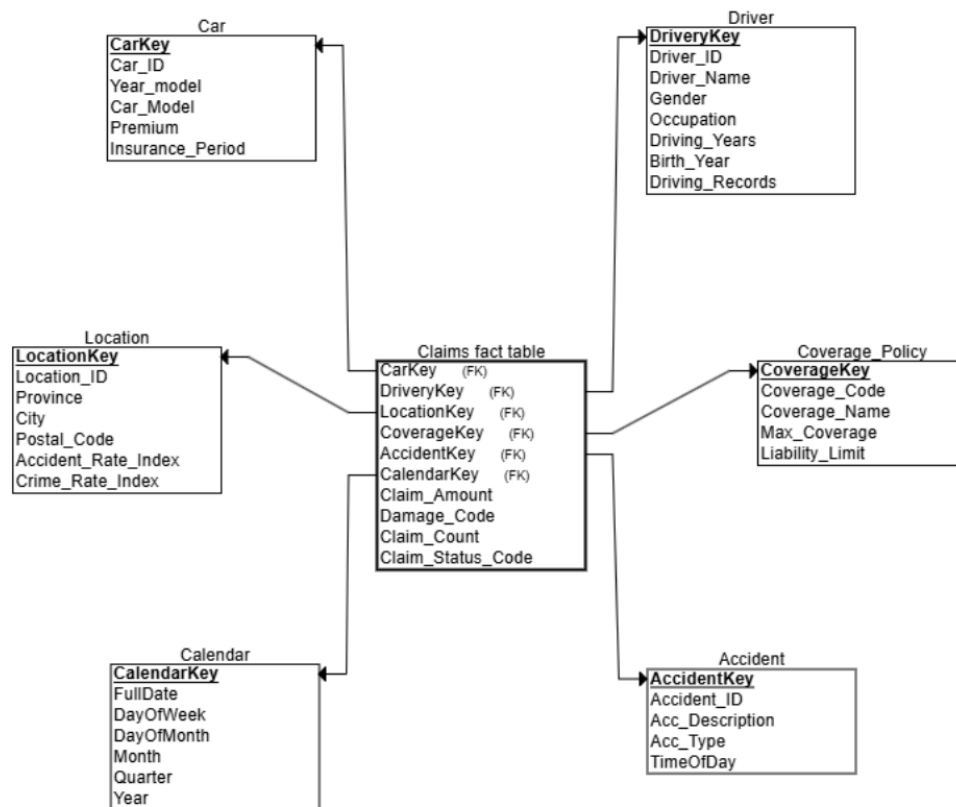
4. Operational Efficiency: By automating data collection and processing, a data warehouse reduces the time and effort required to generate reports and conduct analyses. This efficiency allows insurers to respond swiftly to market changes and emerging risks.

5. Scalability and Flexibility: As the volume of data grows, a well-designed data warehouse can scale accordingly, accommodating new data sources and analytical tools. This flexibility supports ongoing innovation in risk assessment methodologies.

In summary, a data warehouse is indispensable for auto insurance companies aiming to enhance their risk analysis capabilities. It provides a robust foundation for integrating data, performing sophisticated analyses, and making data-driven decisions that improve risk management and competitive advantage.

Task 2:

a) STAR SCHEMA:



b) Star schema for auto insurance is focused on aggregated analysis that often aim to assess metrics like total claims or average premiums across various dimensions.

Measurements in the Fact Table:

The fact table is designed with several key measurements that support comprehensive risk analysis for auto insurance. There were two important questions discussed in the lecture for considering which dimensions to use with the fact table.

Question 1: Can the dimension table be useful for the analysis of the chosen subject?

Question 2: Can the dimension table be created based on the existing data sources?

1. Claim_Amount: This is the primary quantitative measure representing the monetary value of each claim. This metric allows the insurance company to analyze costs across different dimensions and is essential for financial risk assessment.

2. Claim_Count: Although most rows represent a single claim (value of 1), this measure enables aggregation of claim frequency across dimensions. This helps identify patterns of repeated claims that may indicate higher risk factors.

3. Damage_Code: This code categorizes the type of damage, allowing for analysis of claim patterns by damage type.

4. Claim_Status_Code: This status indicator tracks the lifecycle of claims, supporting analysis of processing efficiency and settlement patterns. While it's true that fact tables typically contain numeric measures (like amounts, counts, etc.), they can also contain "degenerate dimensions" that are basically attributes that aren't full dimensions but are still important for analysis. We have used Claim_Status_Code as a degenerate dimension.

Dimensions and Attributes:

For Dimension Driver, we chose to include birth_year with the assumption of Age based analysis. Example, if younger drivers file more claims or premium prices by age group.

1. Car Dimension

Attributes and reasoning: Car_ID, Year_model, Car_Model, Premium and Insurance_Period. The year model and car model helps assess the car's age and car type, which correlates with reliability and repair costs. Premium and insurance period data provide context for financial analysis and policy duration, allowing for return on investment calculations on different vehicle types. As seen during the lecture in class, The Product dimension included ProductVendorName and ProductCategoryName directly from the Vendor and Category table by Denormalizing them. Similarly, we have combined the car and insurance table into one dimension as they have foreign key references.

2. Driver Dimension

Attributes and reasoning: Driver_ID, Driver_Name, Gender, Occupation, Driving_Years, Birth_Year, Driving_Records. Age (derived from birth year), gender, occupation, and driving experience all correlate with accident likelihood. Young drivers may have higher insurance rates. Driving records directly reflect past risk behavior, making this dimension crucial for predictive analysis.

3. Location Dimension

Attributes and reasoning: Location_ID, Province, City, Postal_Code, Accident_Rate_Index, Crime_Rate_Index. The location hierarchy (province, city, postal code) enables regional pattern analysis. The accident and crime rate indices provide contextual risk indicators for each area, supporting better geographic risk assessment. We have calculated them based on per 100 residents.

4. Coverage Policy Dimension

Attributes and reasoning: Coverage_Code, Coverage_Name, Max_Coverage, Liability_Limit. Policy information affects how likely people are to make claims and how much those claims might cost. This dimension helps analyze how different types of coverage and limits are linked to claims, which can help improve insurance plans.

5. Accident Dimension

Attributes and reasoning: Accident_ID, Acc_Description, Acc_Type, TimeOfDay.

Understanding accident circumstances is vital for risk profiling. The type and timing of accidents reveal patterns that can inform preventative measures and pricing strategies. This dimension allows for detailed analysis of what types of accidents generate the highest claims.

6. Calendar Dimension

Selected attributes: FullDate, DayOfWeek, DayOfMonth, Month, Quarter, Year. Time analysis reveals seasonal trends and long-term patterns in claim behavior. The calendar dimension with multiple hierarchical levels (year > quarter > month > day) supports analysis at various granularities, from daily patterns to annual trends.

TASK 3:

Sample Data for Fact and Dimension Tables

Claims Fact Table

CarKey	DriverKey	LocationKey	CoverageKey	AccidentKey	CalendarKey	Claim_Amount	Damage_Code	Claim_Count	Claim_Status_Code
1	3	1	3	2	1	8500	DG3	1	OPEN
2	5	2	2	1	2	3200	DG1	1	CLOSED
4	2	4	4	5	3	12800	DG5	1	PENDING
3	1	3	3	3	4	1950	DG2	1	CLOSED
5	4	5	1	4	5	850	DG1	1	CLOSED
1	3	1	3	2	2	2300	DG2	1	CLOSED
2	5	2	2	5	3	9500	DG4	1	OPEN
3	1	3	3	1	1	4200	DG3	1	CLOSED

Car Dimension Table

CarKey	Car_ID	Year_model	Car_model	Premium	Insurance_Period
1	C1	2021	SUV	1250	12
2	C2	2019	Sedan	1500	6
3	C3	2022	Hatchback	1000	12
4	C4	2018	Van	1580	12
5	C5	2020	Sedan	1250	6

Driver Dimension Table

DriverKey	Driver_ID	Driver_Name	Gender	Occupation	Driving_Years	Birth_Year	Driving_Records
1	D1	Bisrat Mulugeta	M	Engineer	8	1988	Clean title
2	D2	Darshan Jain	M	Doctor	15	1980	1 Minor violations
3	D3	Neha Narendar	F	Accountant	5	1992	Clean title
4	D4	Gemechis Kebe	M	Teacher	3	1995	2 Minor violations
5	D5	Tewodros Tade	M	Manager	12	1984	1 Major violations

Location Dimension Table

LocationKey	Location_ID	Province	City	Postal_Code	Accident_Rate_Index	Crime_Rate_Index
1	L1	Ontario	Toronto	M5V 2B1	0.85	0.78
2	L2	Quebec	Montreal	H2Y 1N9	0.92	0.82
3	L3	Alberta	Calgary	T2P 1J9	0.75	0.65
4	L4	Ontario	Brampton	M5V 3V5	0.88	0.72
5	L5	Manitoba	Winnipeg	R3C 0C4	0.76	0.69

Coverage Policy Dimension Table

CoverageKey	Coverage_Code	Coverage_Name	Max_Coverage	Liability_Limit
1	COV1	Basic	50000	200000
2	COV2	Standard	100000	500000
3	COV3	Premium	250000	1000000
4	COV4	Basic	500000	2000000
5	COV5	Collision Only	75000	300000

Accident Dimension Table

AccidentKey	Accident_ID	Acc_Description	Acc_Type	TimeOfDay
1	A1	Rear-end collision	Collision	8:23:59 AM
2	A2	Side impact	Collision	2:45:23 PM
3	A3	Weather-related skid	Single-car	12:00:09 PM
4	A4	Parking lot scratch	Minor	6:30:00 AM
5	A5	Intersection crash	Multi-car	9:37:40 PM

Calendar Dimension Table

CalendarKey	FullDate	DayOfWeek	DayOfMonth	Month	Quarter	Year
1	15-Jan-2023	Sunday	15	January	Q1	2023
2	22-Feb-2023	Wednesday	22	February	Q1	2023
3	10-May-2023	Wednesday	10	May	Q2	2023
4	03-Aug-2023	Thursday	3	August	Q3	2023
5	27-Nov-2023	Monday	27	November	Q4	2023

TASK 4:

Internal Data Sources:

- Policy Management Systems: Primary source for insurance policy details including coverage types, premiums, and policy periods.
- Claims Processing Systems: Contains detailed information about claims, damage assessments, claim amounts, and settlement processes.
- Customer Relationship Management Systems: Provides driver information, contact details, and historical customer interactions.
- Vehicle Registration Database: Internal records of insured vehicles including model details, year, and value.
- Payment Processing Systems: Contains premium payment records, payment history, and financial transactions.
- Accident and Incident Reports: Stores accident details, including date, location, accident type, and cause.

External Data Sources:

- Government Motor Vehicle Departments: Driver license information, driving records, and vehicle registrations.
- Credit Rating Agencies: Credit scores and financial stability indicators that correlate with risk levels.
- Weather Data Services: Historical weather conditions that might correlate with accident frequencies.
- Statistics Canada: Demographic information, population densities, and broader socioeconomic indicators.
- Police Reports: Detailed accident reports and traffic violation data.
- Postal Code Databases: Geographic information for location-based risk assessment.
- Crime Rate Statistics: Local crime statistics that may affect theft and vandalism risks.
- Traffic Pattern Databases: Information about road usage, traffic densities, and accident hotspots.

Challenges in Building the Data Warehouse:

- Data Integration Complexity: Bringing together data from different sources is hard because they often use different formats, schemas, and updates at different times.
- Data Quality Issues: Inconsistent, incomplete, or inaccurate data requires substantial cleaning and validation processes.
- Historical Data Management: Insurance risk analysis requires historical data, but maintaining and transforming historical records while ensuring consistency is challenging.

- **Privacy and Regulatory Compliance:** Insurance data contains sensitive personal information subject to regulations, requiring careful handling.
- **Real-time vs. Batch Processing:** Determining the right balance between real-time data updates and batch processing for efficient warehouse performance.
- **Scalability Concerns:** Insurance companies process millions of policies and claims, requiring a warehouse design that can scale effectively.
- **Metadata Management:** Maintaining comprehensive metadata to ensure business users understand the meaning and context of warehouse data.
- **Data Granularity Decisions:** Determining the appropriate level of detail to store in the fact table while balancing analytical needs with storage constraints.
- **User Adoption:** Creating a data warehouse that business analysts can effectively use without extensive technical knowledge.

TASK 5:

Risk Analysis: Using the designed data warehouse, an auto insurance company could investigate the following risk analysis questions:

Claim Cost Analysis:

- **Damage Type Analysis:** What types of damages (collision, theft, weather-related) result in the highest average claim costs? This helps prioritize risk mitigation strategies.
- **Geographic Claim Patterns:** How do claim amounts vary by province, city, and postal code? Are there locations with higher than average claim costs that should be assessed differently?
- **Vehicle Profile Impact:** How does the vehicle year and model correlate with claim amounts? Are newer or older vehicles associated with higher claim costs?
- **Seasonal Claim Variations:** Do claim amounts show seasonal patterns? Are winter claims typically more expensive than summer claims?
- **Coverage Level Efficiency:** How do different coverage types compare in terms of claim frequency vs. premium collected? Which coverage types have the best and worst loss ratios?

Driver Risk Profiling

- **Demographic Risk Assessment:** How do factors like age, gender, and occupation correlate with claim frequency and severity?
- **Driving Experience Impact:** What is the relationship between years of driving experience and claim likelihood? At what point does additional experience stop reducing risk?
- **Driving Record Correlation:** How strongly do previous driving records predict future claim likelihood and severity?
- **High-Risk Combinations:** What combinations of driver and vehicle characteristics present the highest risk profiles?
- **Risk Evolution Over Time:** How does a driver's risk profile evolve over time? Do drivers with minor accidents show different long-term patterns than those with clean records?

Location-Based Risk Analysis

- **Accident Hotspot Identification:** Which postal codes have significantly higher accident rates relative to vehicle density?
- **Crime Index Correlation:** How strongly does the local crime rate correlate with vehicle theft and vandalism claims?

- Urban vs. Rural Risk Comparison: How do claim patterns differ between urban and rural locations? Are there significant differences in claim types or amounts?
- Regional Risk Factors: What region-specific risk factors (weather patterns, road conditions, traffic density) contribute to higher claim rates?

Time-Based Analysis

- Time-of-Day Risk Assessment: Which times of day show the highest accident rates and claim severities?
- Day-of-Week Variations: Do accident patterns vary significantly by day of the week? Are weekend claims different in nature from weekday claims?
- Monthly and Seasonal Trends: How do claim frequencies and amounts fluctuate throughout the year? Can these be correlated with specific seasonal factors?
- Long-Term Trend Analysis: Are claim frequencies and severities increasing or decreasing over time? Are certain types of claims becoming more or less common?

Vehicle-Specific Analysis

- Vehicle Safety Feature: What is the correlation between specific vehicle safety features and claim frequency? Which features provide the greatest risk reduction?
- Vehicle Depreciation Impact: How does the depreciation rate of different vehicle models affect the ratio of premiums to potential claim payouts over time?
- Vehicle Age Threshold Analysis: Is there a specific vehicle age at which the risk profile significantly changes? How should premium adjustments reflect this threshold?

These analyses would enable the insurance company to develop more accurate pricing models, implement targeted risk mitigation strategies, and optimize their portfolio management approach.
